

Phase-Based Gravitational Dynamics and Synthetic Metrics: An Epistemic Analysis of Quantum Interferometry and Spatio-Temporal Modulation in the Context of UAT/UPC

Miguel A. Percudani

Independent Researcher, Argentina

ORCID: 0009-0007-1748-3212

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Abstract

Recent experimental developments in quantum interferometry and spatio-temporally modulated resonant systems provide important empirical context for phase-based approaches to physical dynamics. In particular, the direct measurement of quantum phase accumulation during free fall reported by Dobkowski et al. [6], and the experimental realization of magnetic-free non-reciprocity through parametric phase modulation reported by Estep, Sounas, Soric, and Alù [5], demonstrate that phase evolution and its controlled spatio-temporal modulation can constitute experimentally observable physical degrees of freedom.

These results do *not* validate the Universal Applied Time (UAT) or Unified Principle of Causality (UPC) frameworks. Both experiments were developed and interpreted within established physical frameworks and address specific experimental questions unrelated to the validation of UAT/UPC. Nevertheless, they provide relevant experimental precedents for a research strategy in which discrete phase geometry, temporal modulation, and coherent superposition are treated as primary experimental variables.

Within this context, the present work identifies the eight-coil rotational matrix developed under the UAT/UPC program as an exploratory macroscopic platform for testing whether controlled phase geometry can produce reproducible collective observables. The objective is not to assume the existence of a new metric or causal field, but to determine experimentally whether a reproducible residual or convergence parameter emerges from an independently calibrated phase-rotational system.

1 Introduction

The experimental investigation of gravitational phenomena at the quantum level increasingly emphasizes the role of phase as an observable quantity. In an interferometric experiment, gravitational dynamics can be encoded in the relative phase accumulated by distinct quantum trajectories. This provides an experimentally accessible description in which gravitational effects appear through phase evolution rather than solely through a classical force observable.

A particularly relevant recent result is the direct measurement of the quantum phase associated with free fall reported by Dobkowski et al. [6]. Their experiment employs a Quantum Galileo Interferometer (QGI) to measure the differential phase accumulated between a stationary and a freely falling component of an atomic wave packet.

The experiment provides an important benchmark for phase-based descriptions of gravitational dynamics. In particular, the observed time dependence contains a cubic contribution of the form

$$\phi_{\text{ff}} \propto T^3, \quad (1)$$

consistent with the theoretical phase evolution expected for the free-fall configuration investigated in that experiment.

This result should not be interpreted as evidence for UAT or UPC. Rather, it establishes an experimentally demonstrated example in which gravitational dynamics are directly encoded in a measurable quantum phase.

The distinction is essential:

experimental evidence for gravitational phase evolution $\neq$ experimental validation of UAT/UPC

(2)

The significance for the present research program is therefore methodological rather than confirmatory.

2 The Alù Precedent: Spatio-Temporal Phase Modulation

A second relevant experimental precedent is provided by Estep, Sounas, Soric, and Alù [5], who demonstrated magnetic-free non-reciprocity in a system consisting of three strongly coupled resonators whose resonance frequencies were parametrically modulated.

The three modulation signals were phase shifted by

$$\Delta\phi = \frac{2\pi}{3} = 120^\circ. \quad (3)$$

The resulting spatio-temporal modulation generated an effective angular-momentum bias in the coupled-resonator system and produced strong non-reciprocal transmission.

The experimental result establishes an important general principle: the relative temporal phase of spatially distributed resonant elements can modify the collective dynamical response of a physical system.

This can be represented schematically as

$$\text{distributed resonators} + \text{temporal phase modulation} \longrightarrow \text{modified collective dynamics}. \quad (4)$$

The Alù experiment does not demonstrate gravitational dynamics, synthetic space-time curvature, UAT, or UPC. Its physical system and experimental objective are different. Nevertheless, it provides a well-established experimental precedent for treating phase geometry and temporal modulation as active variables in a distributed physical system.

This distinction is central to the present work.

3 From Three-Element Phase Modulation to an Eight-Coil Matrix

The present experimental program explores a related but distinct configuration based on eight phase-coherent elements.

For an ideal eight-element rotational phase structure, the discrete phase increment is

$$\Delta\phi = \frac{2\pi}{8} = \frac{\pi}{4} = 45^\circ. \quad (5)$$

The corresponding phase sequence may be represented as

$$\phi_k = \phi_0 + k\frac{\pi}{4}, \quad k = 0, \dots, 7. \quad (6)$$

The eight-coil configuration is therefore not based on eight coils being physically opposed to one another. Instead, the rotational relationship can be encoded through the relative phase and frequency of the individual channels.

This distinction is experimentally important.

The physical arrangement can remain comparatively simple while the rotational geometry is implemented in the signal domain:

$$\boxed{\text{physical coil array} + \text{controlled phase relationships} = \text{synthetic rotational configuration}}. \quad (7)$$

The eight-coil detector and its calibration package are documented in Percudani [4], including the executable calibration code, detector configuration, calibration methodology, and final parameter set.

4 The Eight-Coil Rotational Matrix

The eight-coil system was developed as an exploratory experimental platform within the UAT/UPC research program.

Its principal objective is to determine whether a controlled rotational phase configuration produces a reproducible collective observable after the individual channel contributions are combined.

The relevant phase structure is

$$\{0^\circ, 45^\circ, 90^\circ, 135^\circ, 180^\circ, 225^\circ, 270^\circ, 315^\circ\}. \quad (8)$$

For eight pairwise channel relationships, the number of unordered pairs is

$$\binom{8}{2} = \frac{8!}{2!6!} = 28. \quad (9)$$

The value 28 is also the second perfect number in classical number theory. This numerical correspondence is noted here as a structural property of the eight-element configuration and is not, by itself, interpreted as evidence of a physical law.

5 Rotational Superposition and Residual Signal

For an idealized set of eight equal-amplitude phase components, the complex rotational sum is

$$S_8 = \sum_{k=0}^7 A e^{i(\phi_0 + k\pi/4)}. \quad (10)$$

Using the geometric-series identity,

$$S_8 = A e^{i\phi_0} \sum_{k=0}^7 e^{ik\pi/4}. \quad (11)$$

Since

$$e^{i8\pi/4} = e^{i2\pi} = 1, \quad (12)$$

the ideal symmetric sum satisfies

$$\boxed{S_8 = 0}. \quad (13)$$

Consequently, any experimentally measured residual,

$$S_{\text{res}} \neq 0, \quad (14)$$

requires further analysis.

Possible contributions include

$$S_{\text{res}} = S_{\text{instrumental}} + S_{\text{phase}} + S_{\text{frequency}} + S_{\text{electronic}} + S_{\text{physical}}. \quad (15)$$

The central experimental question is therefore not whether a residual can be produced, but whether a residual remains reproducible after instrumental, electronic, phase, and frequency-dependent effects have been independently characterized.

6 The Observed Residual and the Candidate Invariant

During the development of the eight-coil rotational configuration, a residual value close to

$$\boxed{R \simeq 0.2971} \quad (16)$$

was identified under the corresponding phase/frequency confrontation procedure.

This value is treated here as a *candidate invariant*, not as an established universal constant.

The distinction is essential. A numerical residual obtained from a single configuration does not establish universality. To support such an interpretation, the quantity must remain stable under controlled variations of parameters that should not affect the proposed invariant.

A suitable experimental program would therefore measure

$$R = R(f, A, \phi_0, \Delta\phi, T, G, \dots), \quad (17)$$

where the variables represent frequency, amplitude, initial phase, phase increment, integration time, gain, and other relevant instrumental parameters.

A stronger indication of invariance would occur if

$$R(f, A, \phi_0, \Delta\phi, T, G, \dots) \approx 0.2971 \quad (18)$$

over a statistically significant region of parameter space, while remaining distinguishable from the instrumental noise floor.

7 The RMS Coherence Scale

The eight-coil calibration study also identified a central-coil RMS value close to

$$RMS_0 \simeq 0.707001, \quad (19)$$

which is close to the theoretical value

$$\frac{1}{\sqrt{2}} = 0.707106781\dots \quad (20)$$

reported as the relevant causal-buffer scale in the UAT/UPC calibration framework [4].

This quantity and the candidate residual

$$R \simeq 0.2971 \quad (21)$$

must not be assumed to represent the same physical observable.

A complete analysis must explicitly define the mathematical operator that generates each quantity and determine whether any relationship between them follows from the underlying model rather than from calibration choices.

8 Conceptual Convergence

The significance of the recent experimental literature for the present research program can therefore be formulated as a *conceptual convergence*, rather than as a direct validation.

The relevant empirical results can be summarized as follows:

$$\begin{aligned} \text{Dobkowski et al.} &\longrightarrow \text{direct measurement of gravitational quantum phase,} \\ \text{Estep et al./Alù} &\longrightarrow \text{controlled spatio-temporal phase modulation,} \end{aligned} \quad (22)$$

8-coil UAT/UPC experiment $\longrightarrow$ test of collective rotational phase coherence.

These three lines of work address different physical systems and different scientific questions. Accordingly,

$$\boxed{\text{Dobkowski et al. and Alù et al. do not validate UAT/UPC}} \quad (23)$$

and no such inference is made in this work.

Their relevance is instead that they independently demonstrate that phase accumulation and controlled spatio-temporal phase relationships can be experimentally measured and can produce non-trivial collective physical responses.

This provides experimental motivation for continuing to investigate phase-based architectures such as the eight-coil rotational matrix.

9 Epistemic Boundaries and Falsifiability

The UAT/UPC interpretation remains a hypothesis requiring independent experimental validation.

In particular, the following statements are deliberately kept separate:

1. Quantum gravitational phase accumulation has been experimentally measured in an established theoretical framework.
2. Spatio-temporal phase modulation has experimentally produced non-reciprocal dynamics in coupled resonator systems.
3. The eight-coil rotational configuration provides an experimentally testable implementation of discrete phase relationships.
4. A residual near 0.2971 has been observed within the corresponding experimental analysis.
5. The physical universality of 0.2971 remains an open experimental question.
6. None of the preceding results, individually or collectively, constitutes direct experimental validation of UAT or UPC.

This separation is essential for maintaining a falsifiable research program.

10 Experimental Objective

The immediate experimental objective is therefore not to demonstrate UAT/UPC by assumption, but to determine whether the eight-coil rotational matrix exhibits a statistically reproducible invariant under controlled experimental variation.

The central hypothesis can be written as

$H_0 : R \neq 0$ is entirely explained by instrumental and systematic effects, (24)
against the alternative

$H_1 : R \approx 0.2971$ persists after independently characterized instrumental effects are removed. (25)

The distinction between these hypotheses provides a direct path toward experimental falsification.

If the residual disappears under improved shielding, independent instrumentation, phase randomization, frequency variation, or configuration reversal, its origin can be attributed to the relevant systematic effect.

If, instead, the residual remains stable under such controls, further investigation would be justified.

11 Conclusion

Recent experimental developments in quantum interferometry and spatio-temporal modulation have established important empirical facts about the physical role of phase.

The direct measurement of gravitational quantum phase by Dobkowski et al. [6] demonstrates that gravitational free-fall dynamics can be directly accessed through quantum phase interferometry.

The experiment of Estep, Sounas, Soric, and Alù [5] demonstrates that controlled temporal phase modulation in a discrete resonator system can produce a measurable modification of collective dynamics and non-reciprocal response.

Neither result validates UAT or UPC.

Nevertheless, together they establish an experimentally grounded context in which phase geometry, temporal modulation, and coherent superposition can be treated as legitimate experimental variables.

The eight-coil rotational matrix represents a distinct attempt to investigate whether a related phase-structured architecture produces a reproducible collective residual or invariant.

The candidate value

$$R \simeq 0.2971 \quad (26)$$

therefore remains an empirical hypothesis rather than an established universal constant.

Its scientific significance will ultimately depend on whether it can be reproduced independently, survives appropriate instrumental controls, and remains invariant under controlled changes in the experimental parameters.

The appropriate conclusion at the present stage is consequently not that UAT/UPC has been confirmed, but that the experimental route based on phase geometry and controlled temporal modulation has acquired stronger independent physical precedents than were available when the eight-coil hypothesis was initially formulated.

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